

$Z \rightarrow \mu\mu$ CROSS SECTION: REVIEW OF THE V2 MEASUREMENT

CMS 2016 Open Data, Run2016G+H, 16.39 fb^{-1} – 15 September 2026

Result under review: $\sigma_{\text{fid}}(Z/\gamma^* \rightarrow \mu\mu) = 791.8 \pm 0.2 \text{ (stat)} \pm 6.2 \text{ (syst)} \pm 9.3 \text{ (lumi) pb}$, $\mu_Z = 0.990 \pm 0.014$ from a 30-bin TRExFitter fit of $m_{\mu\mu}$.

Verdicts

- $e\mu$ region excess (+9 %): the non-prompt-electron background that the prompt-only MC filter removes. Put back: data/MC = 0.996. Validation region only, no effect on σ_{fid} .
- p_T^{miss} in the SR: unsmeared MC jets + pileup/unclustered energy. Puppi p_T^{miss} in 0-jet events agrees to 1 %. Not used anywhere, no effect.
- **The fit is not robust:** $\mu_Z = 0.990 / 0.996 / 0.9935 / 1.005$ with binning and shape templates; a 3–4 % data deficit at 62–78 GeV is absorbed by over-constrained shape NPs. Recommend the counting extraction $(794.4 \text{ pb}) \pm 0.7 \%$ until fixed.
- Sound: tight ID, dressed fiducial volume, normtag luminosity, T&P fits, fake factor, momentum calibration, NLO acceptance.

Sections

1. Result and the fit
2. The $e\mu$ control region
3. p_T^{miss} in the signal region
4. Analysis strategy and inputs
5. Choices reviewed, recommendations
6. Backup

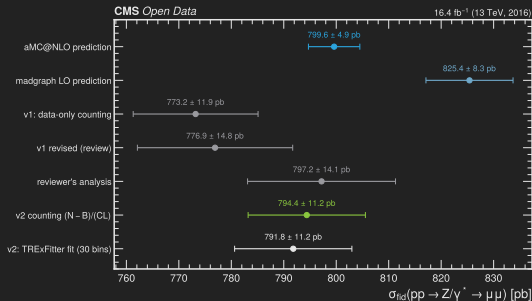
Full write-up: `z-mumu/REVIEW.md`

figures: `z-mumu/review/figures/`

fit variants: `z-mumu/review/fitcheck/`

#	finding	status	effect on σ_{fid}
F1	$e\mu$ region +9.2%: non-prompt electrons (W +jets $\text{jet} \rightarrow e$, $Z \rightarrow \mu\mu + \gamma$ conversions, $Z \rightarrow \tau\tau$ $\tau_h \rightarrow e$, $t\bar{t}$ $b \rightarrow e$) removed from MC, not replaced	understood	none (validation region)
F2	p_T^{miss} broader in data: no JER smearing of MC jets; pileup/unclustered energy; Z p_T modelling	understood	none (not used)
F3	μ_Z moves by up to 1.5% with binning / shape templates; only the nominal configuration has an acceptable GoF	must fix	up to $\pm 0.8\%$
F4	powheg SigModel template mixes a generator-cut artefact ($m_{\text{LHE}} < 120$: -33% in the last bin) with the generator comparison	must fix	part of F3
F5	NPs constrained to 0.1–0.2 of their prior (MuonScale, SigModel, MuonRes); Lumi constrained to 0.82 by the $t\bar{t}$ sideband; Pileup pulled -1.5σ the wrong way	should fix	lumi 1.16% $\rightarrow$ 1.20%
F6	pileup profile $\approx 4\%$ too low: data/MC in N_{PV} rises 0.8 $\rightarrow$ 1.2	should fix	$\leq 0.05\%$
F7	jet multiplicity not lepton-cleaned (plots peak at 2 jets)	bug (plots)	none
F8	MuonReco: 0.4% per muon in the docs, 0.4% per event in the fit	inconsistent	+0.35% if per muon
F9	MC statistics $1.7 \times$ the data statistics per bin; gammas pulled up to 1.3σ	inherent	0.35% (quoted)

1. RESULT AND THE FIT



quantity	value
σ_{fid} (fit, 30 bins)	791.8 ± 11.2 pb
σ_{fid} (counting $(N - B)/(CL)$)	794.4 pb
μ_Z	$0.990^{+0.014}_{-0.014}$
$\sigma(60 < m < 120), A = 0.4092$	1935 ± 30 pb
$\sigma(m > 50), A = 0.3947$	2006 ± 31 pb
C (reco / fiducial)	0.7917
aMC@NLO $\sigma_{\text{fid}}^{\text{pred}}$	799.6 pb
external reviewer, same volume	797.2 pb

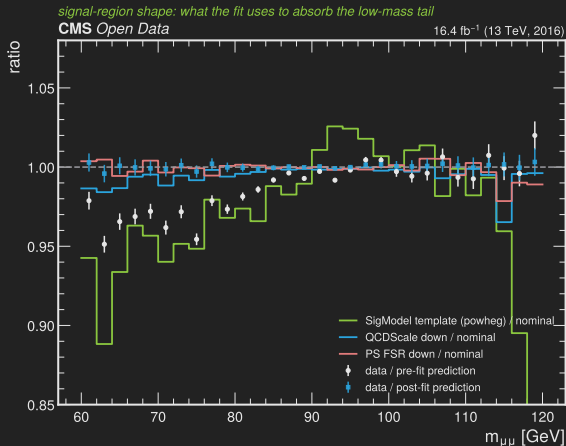
- Every extraction lands within 1 % of the aMC@NLO prediction; the physics is fine.
- The 0.03 % “statistical” uncertainty is misleading: the effective statistical limit is the MC (0.35 %).
- The counting result is highlighted as the robust number (next slides).

THE FIT IS NOT ROBUST: μ_Z VERSUS FIT CONFIGURATION

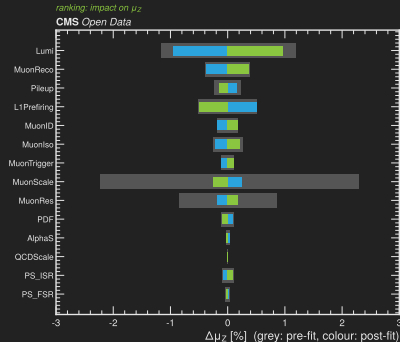
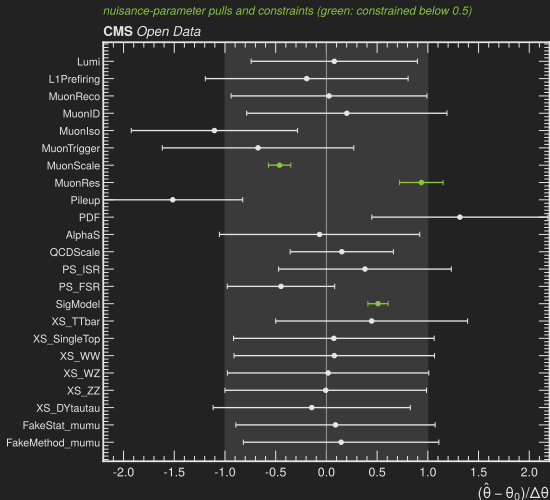
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configuration (review/fitcheck/)	μ_Z	GoF p	notable pulls (constraints)
nominal: 30×2 GeV	$0.9903^{+0.0141}_{-0.0138}$	0.33	SigModel +0.5 (0.10), MuonScale (0.11), MuonRes (0.21), Pileup -1.5 , PDF +1.3, MuonIso -1.1
6×10 GeV	0.9959 ± 0.0160	0.0009	MuonIso -1.7 , PS_ISR +1.3, PS_FSR +1.2
1 bin (= counting)	0.9935	–	grouped total 1.49%
30 bins, no SigModel	$1.0050^{+0.0141}_{-0.0137}$	0.0014	MuonIso -3.0σ , QCDscale -0.7 , MuonScale (0.06)
30 bins, no SigModel, no smoothing	$1.0063^{+0.0141}_{-0.0137}$	0.0003	MuonIso -3.1σ , Pileup (0.48), Lumi (0.22)

- Spread $0.990-1.006 = \pm 0.8\%$, larger than the whole non-lumi systematic (0.77%). In σ_{fid} : 792 to 805 pb.
- Data show a 3–4 % deficit against aMC@NLO at 62–78 GeV. With the powheg template the fit explains it as “generator”; without it, as an isolation efficiency 1 % below tag-and-probe. The data cannot tell which.
- 10 M events in 2 GeV bins resolve 10^{-3} shape differences: finer than the MC precision of the templates and than their physics content. The resulting constraints are artefacts (Lumi measured by the $t\bar{t}$ sideband shape).
- The counting extraction is insensitive to the lineshape to first order: C is a ratio of the same MC in the same window.



- Grey: data / pre-fit prediction. 0.95–0.97 for $62 < m < 78$ GeV, 1.00 ± 0.005 above 86 GeV. Backgrounds are 7 % of the first bin and validated by the $e\mu$ region: this is the Z/γ^* lineshape (continuum, FSR tail, resolution).
- Green: powheg / aMC@NLO. –6 to –11 % at 60–70 GeV, +2.5 % at the peak, –10 % / –33 % in the last two bins from the $m_{\text{LHE}} < 120$ generator cut of the powheg sample (resolution migration). That edge is a phase-space artefact, not modelling.
- QCD-scale and PS-FSR variations (blue, red) are flat at the 0.5 % level: no other NP can describe the tail.
- Blue points: data / post-fit, flat, $p = 0.33$, obtained with SigModel pulled $+0.5\sigma$ and constrained to 0.10.



Pre-fit impacts (grey) of MuonScale (2.3%) and SigModel (2.8%) shrink to 0.25–0.3% post-fit: the 2 GeV shape “measures” them. Pileup is pulled towards *less* pileup while N_{PV} asks for more (F6): the mass shape carries no pileup information.

UNCERTAINTY BUDGET (AS QUOTED) AND WHAT THE REVIEW CHANGES

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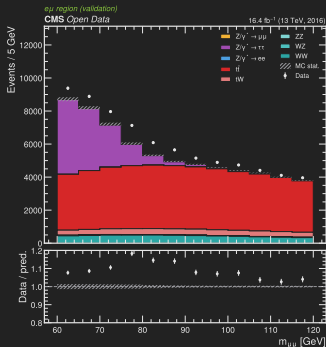
group	impact on μ_Z	review
luminosity	1.16 %	quote 1.20 % (F5)
L1 prefiring	0.51 %	ok
muon efficiency	0.50 %	F8: +0.35 %?
signal modelling	0.36 %	F3/F4: +0.7 %
MC statistics	0.35 %	inherent
muon momentum	0.27 %	calibration used twice
pileup	0.16 %	F6 profile
backgrounds	0.09 %	ok
fakes	0.05 %	ok
statistical	0.03 %	ok
total	1.40 %	≈ 1.6 % after fixes

- MC statistics per mass bin is $1.7\times$ the data statistics: the 30 gammas are pulled up to 1.3σ and constrained to 0.6. The fit is MC-limited, not data-limited.
- A lineshape-model term of ± 0.7 % (half the spread of the fit variants) should be added until the 62–78 GeV deficit is understood.
- MuonReco: docs say 0.4 % per muon (v1 used 0.8 % per event), the fit applies 0.4 % per event.
- The acceptance uncertainty (0.61 %: PDF 0.52, scales 0.32) is correctly kept outside σ_{fid} and added only for σ_{tot} .

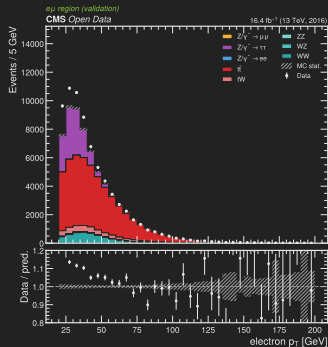
2. THE E_μ CONTROL REGION

THE PROBLEM AS PLOTTED: +9 % IN THE OPPOSITE-SIGN $e\mu$ REGION

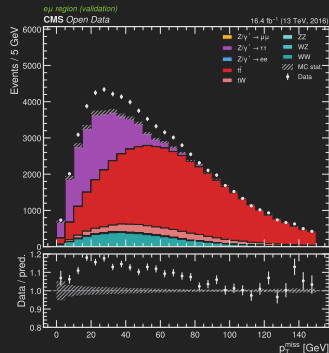
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$m_{e\mu}$: excess at 75–90 GeV



electron p_T : 1.27 at 20–25 GeV, 1.02 above 55 GeV

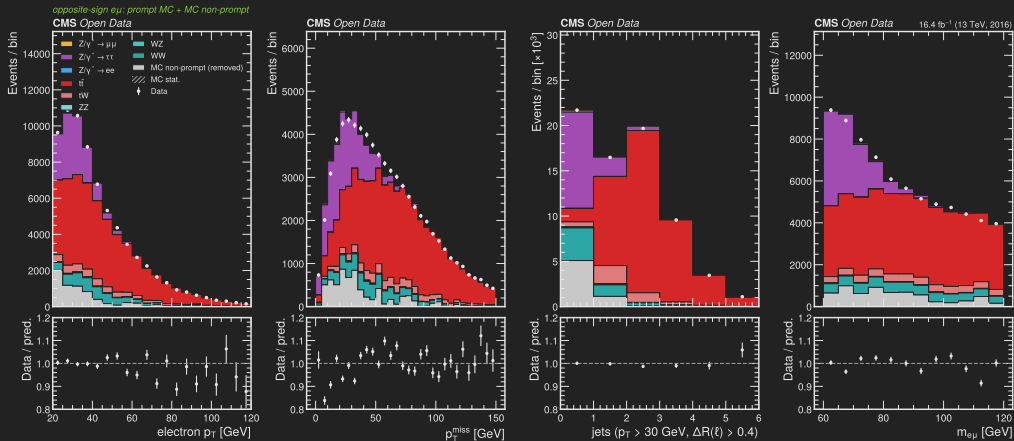


p_T^{miss} : excess at 10–50 GeV

Low electron p_T , no jets (next slide), moderate p_T^{miss} , visible mass just above $Z \rightarrow \tau\tau \rightarrow e\mu$: the signature of a **non-prompt electron** next to a real W/Z muon, not a normalisation problem of $t\bar{t}$ or WW (which populate ≥ 2 jets and agree to 1%).

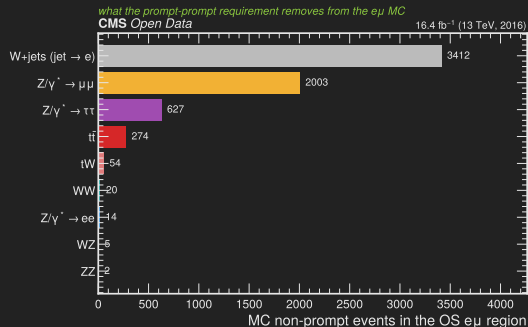
THE EXPLANATION: THE PROMPT-ONLY MC FILTER REMOVES A WHOLE BACKGROUND

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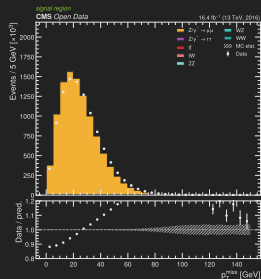
histograms_fill_chunk keeps MC events only if both leptons have $\text{genPartFlav} \in \{1, 15\}$. In the SR the fake factor replaces the removed muons; in the $e\mu$ region nothing replaces the removed electrons. Grey = what was removed (review pass over the skims), now included: data/MC = 0.996, flat in every variable.

opposite-sign $e\mu$	events
data	72 357
prompt-prompt MC (as in v2)	66 232
data — prompt MC	6 125 (+9.2%)
MC non-prompt, removed: W+jets 3 412, $Z \rightarrow \mu\mu + \gamma$ 1 344, $Z \rightarrow \mu\mu$ other 660, $Z \rightarrow \tau\tau$ $\tau_h \rightarrow e$ 627, top 328, VV 27	6 412
data / (prompt + non-prompt)	0.996
same-sign $e\mu$ (new region)	events
data	6 718
prompt-prompt MC	1 039
charge-symmetric non-prompt (data-driven)	5 679
+ OS-only ($\tau_h \rightarrow e, b \rightarrow e$ from MC)	955
SS-based prediction for the OS excess	6 630

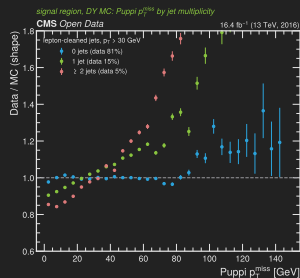
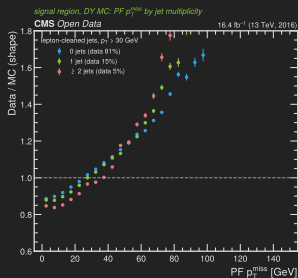


Consequences: none for σ_{fid} (the region is VALIDATION). But $-\mu_{\text{toP}}$ would absorb 9% of non-prompt events into $t\bar{t}$: keep it a validation region until the non-prompt component is included. Electron scale factors would move the prediction the *wrong* way (0.97–0.99).

3. p_T^{miss} IN THE SIGNAL REGION

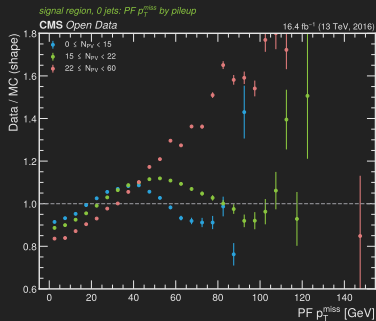
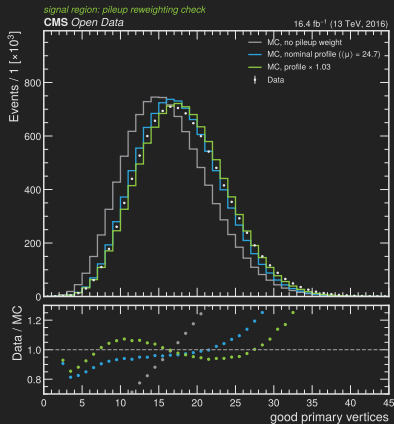


PF p_T^{miss} in the SR: mean 26.5 (data) vs 25.1 GeV (MC)



data/MC shape by lepton-cleaned jet multiplicity: PF (left) and Puppi (right)

- $Z \rightarrow \mu\mu$ has no genuine p_T^{miss} : this is a **resolution** difference. Puppi p_T^{miss} in 0-jet events (81 % of the SR) agrees to $< 1\%$ in every bin.
- The residual grows with the number of jets for PF and Puppi alike: MC jets are not JER-smeared (NanoAOD MC, no smearing applied), so the recoil is too well measured.
- PF p_T^{miss} additionally worsens with pileup (next slide): pileup / unclustered energy, and the profile of F6. Z p_T mismodelling at low p_T (0.83 at 0–5 GeV) adds to it.
- p_T^{miss} is not used in the selection or the fit: **no effect** on σ_{fid} . If ever cut on: JER smearing or Puppi p_T^{miss} , plus MET_Unclustered.



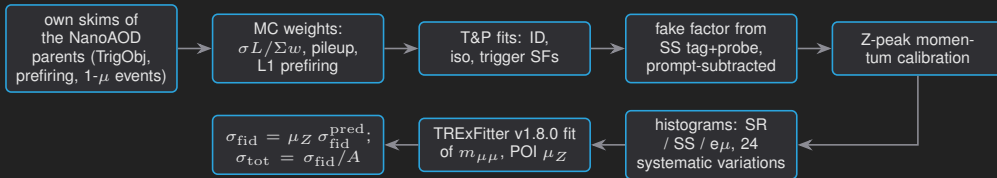
	$\langle N_{PV} \rangle$
data	18.22
MC, no weight	16.26
MC, nominal	17.72
era G / H	17.7 / 18.7

MC:

$N_{PV} = 2.16 + 0.63 N_{\text{true}}$,
so data want $\langle N_{\text{true}} \rangle \approx 25.6$
vs 24.6 after reweighting.

The profile from the per-lumisection avgpu (69.2/80 scaling) lacks the bunch-to-bunch spread and is ~ 1 interaction low. Effect on $\sigma_{\text{fid}} \leq 0.05\%$ (isolation SF measured in data, flat in N_{PV} to 0.5%). The fit's Pileup pull (-1.5σ , towards less pileup) contradicts this: it is template noise, not a measurement.

4. ANALYSIS STRATEGY AND INPUTS

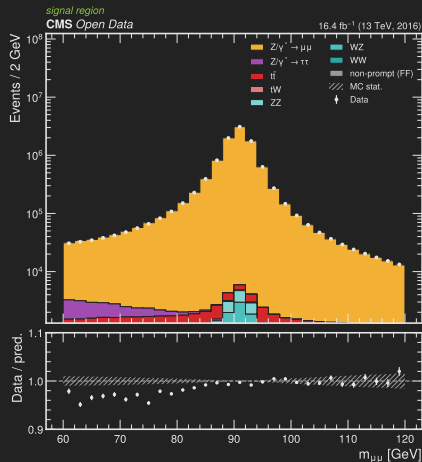
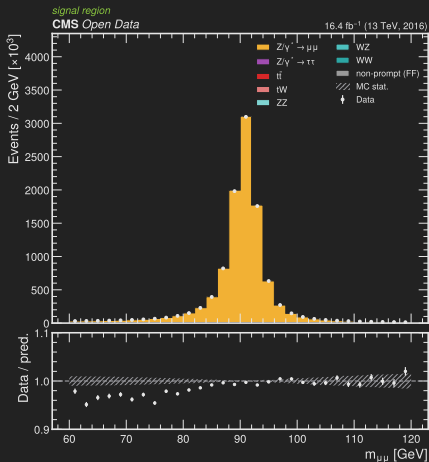


Selection (signal region)

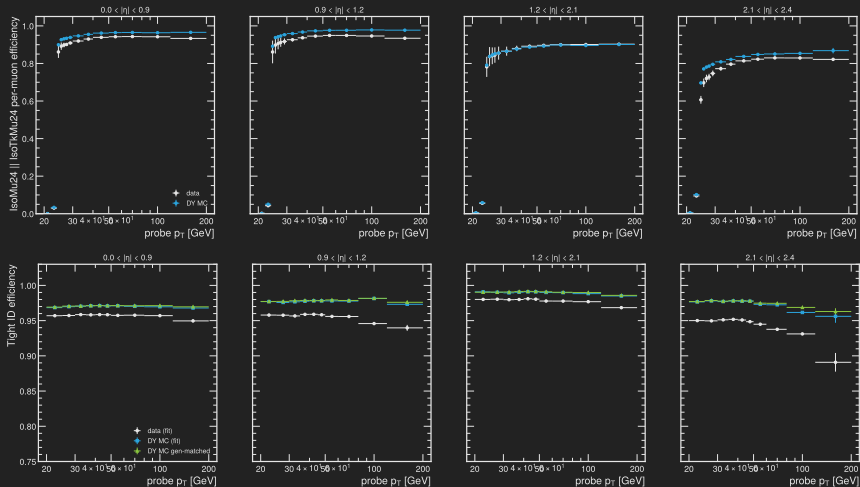
- HLT_IsoMu24 || HLT_IsoTkMu24, MET filters, $N_{PV} \geq 1$
- exactly two tight muons (tightId, iso < 0.15, $|d_{xy}| < 0.2$, $|d_z| < 0.5$ cm), $p_T > 26/20$, $|\eta| < 2.4$, ≥ 1 trigger-matched, opposite sign
- FSR-recovered $60 < m_{\mu\mu} < 120$ GeV; 30 bins of 2 GeV

Fiducial volume and normalisation

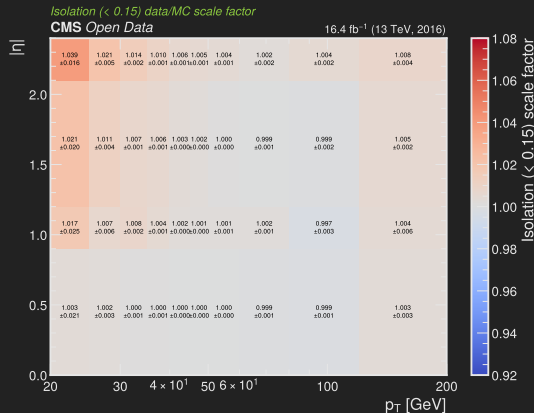
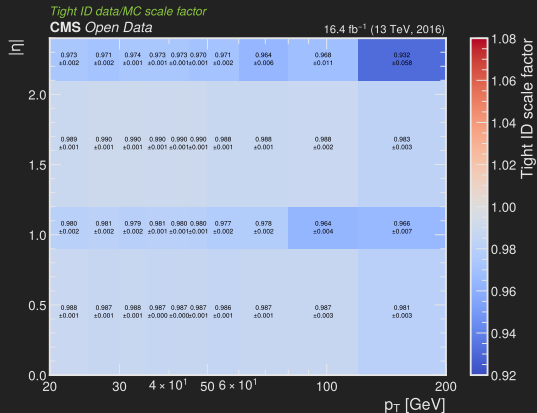
- dressed muons ($\Delta R < 0.1$), $p_T > 26/20$, $|\eta| < 2.4$, $60 < m < 120$ GeV; $C = 0.7917$, $A_{60-120} = 0.4092$ (aMC@NLO)
- $\sigma(Z/\gamma^* \rightarrow \ell\ell, m > 50) = 6077.22/3$ pb, $L = 16393.381$ pb⁻¹ (normtag)
- backgrounds: $Z \rightarrow \tau\tau$, $t\bar{t}$, tW , WW , WZ , ZZ from MC (0.6%), non-prompt from data (0.035%)



10378567 data events; prompt MC + fake factor 10445834 pre-fit. The 3–4% deficit at 62–78 GeV (left ratio) is what drives F3.



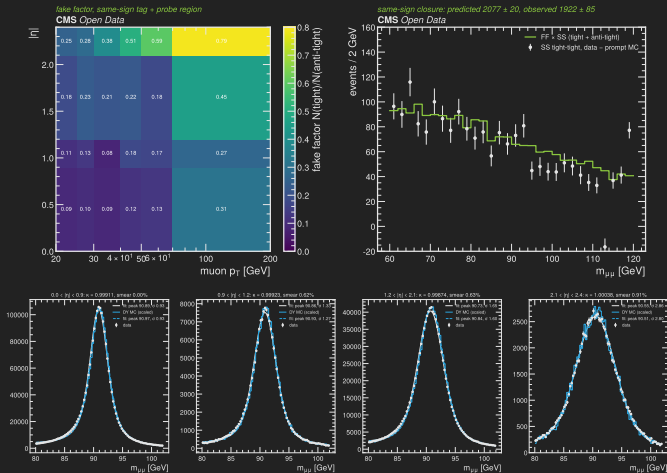
Trigger-object T&P (top; plateau 0.907 data / 0.923 MC per muon) and tight-ID pass/fail fits (bottom; closure to gen-matched truth —0.1 %). Sound.



tight ID: 0.97–0.99, $\langle SF \rangle = 0.980 \pm 0.0014$ (stat) ± 0.0029 (syst)

isolation: 1.00–1.04, $\langle SF \rangle = 1.005$; flat in pileup (half-spread 0.0024)

The MuonIso variation is applied coherently to all cells; its shape (+1.1 % at 60 GeV, +0.2 % at 120 GeV) is what the fit exploits when SigModel is absent (F3).



Fake factor (SS tag+probe, prompt-subtracted): SR non-prompt 3825 ± 80 (stat) $\pm 30\%$ (method), closure 2077 ± 20 vs 1922 ± 85 . Momentum: κ and extra smearing per $|\eta|$; then used again as MuonScale/MuonRes in the fit.

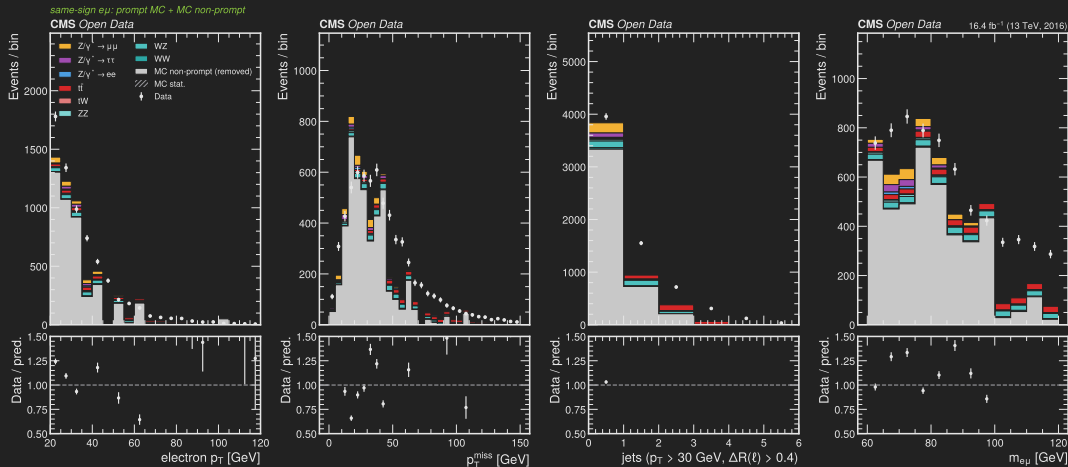
5. CHOICES REVIEWED, RECOMMENDATIONS

choice	verdict	comment
tight ID + iso < 0.15, IP cuts; dressed fiducial volume	good	matches the external reference; dressed avoids the 1.8 % Born ambiguity
own skims; normtag luminosity; L1 prefiring weight	good	prefiring not double counted by the T&P (prefired events never enter)
T&P with pass/fail fits, trigger-matched tag, per-event trigger weight	good	closure $-0.1/-0.2\%$; $\chi^2/\text{ndf} \approx 2$ in data covered by variants
fake factor from SS tag+probe; single- μ +jet region rejected	good	template fit handles the Z-dominated anti-isolated sample
Z-peak momentum calibration	good	but the same peak then constrains the NPs to 0.1σ (used twice)
pileup profile from the luminosity CSV	stop-gap	4 % low, no bunch spread (F6)
prompt-prompt MC in SR/SS (0.006 % loss, FF replaces)	good	
prompt-prompt MC in the $e\mu$ region, no replacement	wrong	F1
theory variations renormalised to constant fiducial yield	good	correct for a fiducial measurement
30×2 GeV shape fit with 24 NPs + 30 gammas	not robust	F3/F5
powheg SigModel one-sided template	wrong as built	F4
$e\mu$ as VALIDATION; NLO acceptance, 60–120 denominator	good	keep VALIDATION until F1 is fixed
jets not lepton-cleaned in the plots	bug	F7

1. **Signal extraction (F3–F5).** Rebuild `SigModel` with the same LHE mass window on both generators; understand the 62–78 GeV deficit (generator-level lineshapes, unfolded data); until then quote the counting extraction (794.4 pb) with a $\pm 0.7\%$ lineshape term, keep the 30-bin fit as a cross-check. If a shape fit is kept: coarse bins, consistent smoothing, external `Lumi` (1.2%), pre-fit values for `Pileup`/`MuonIso`.
2. **$e\mu$ region (F1).** Keep the non-prompt MC or use the same-sign estimate; add the SS $e\mu$ region to `regions.py` and the plots. Then the flavour-symmetric backgrounds are validated to $\sim 2\%$ (0.01% on σ_{fid}).
3. **`MuonReco` (F8).** Per muon or per event: fix the config or the documentation (+0.35% in quadrature if per muon).
4. **`Pileup` (F6).** N_{PV} -matched profile or the official JSON; re-check the `Pileup` pull.
5. **`Jets` (F7).** $\Delta R(\text{jet}, \ell) > 0.4$ cleaning in `histograms.py`.
6. p_T^{miss} : nothing for $\mu\mu$; JER smearing or Puppi $p_T^{\text{miss}} + \text{MET_Unclustered}$ wherever p_T^{miss} is cut on (relevant for the $\tau\tau$ channel, which shares these conventions).
7. Z p_T reweighting or an explicit NP (enters C via the p_T thresholds at 0.1–0.2%).

For the combination: use $\sigma_{\text{fid}} = 794.4$ pb (counting) or the fit value with the extra 0.7% until item 1 is done; the correlated NP names and the acceptance convention (NLO, $60 < m < 120$) are fine as they stand.

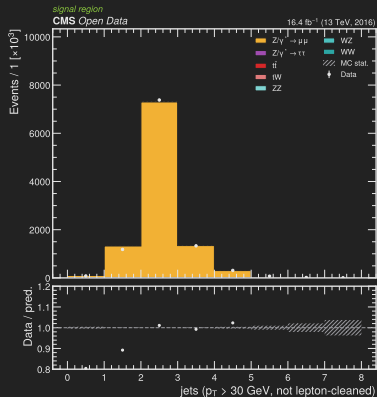
6. BACKUP



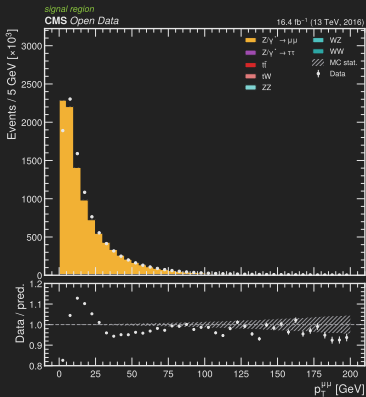
Same-sign $e\mu$: 6 718 data, 1 039 prompt MC, 4 275 MC non-prompt. The MC underestimates the SS region by 26% (no QCD sample, tiny W+jets skim): the data-driven SS number is the reliable one.

BACKUP: THE JET-MULTIPLICITY BUG AND OTHER SR DISTRIBUTIONS

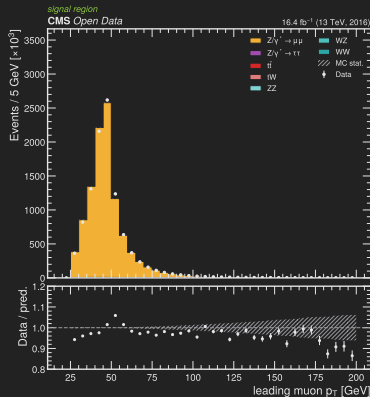
27



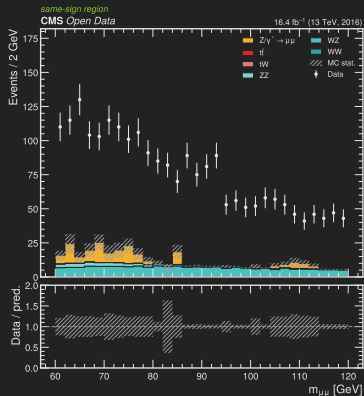
"jets" peak at 2: the two muons are counted (F7)



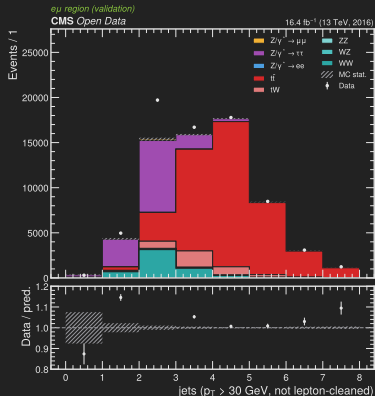
$Z p_T$: 0.83 at 0–5 GeV, 1.13 at 10–15 GeV



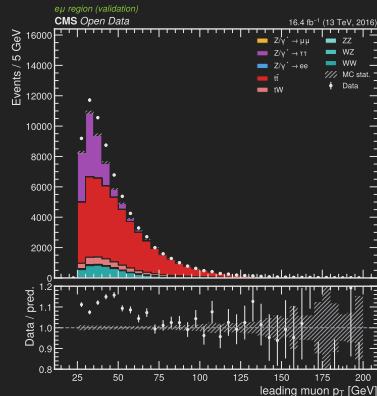
leading muon p_T



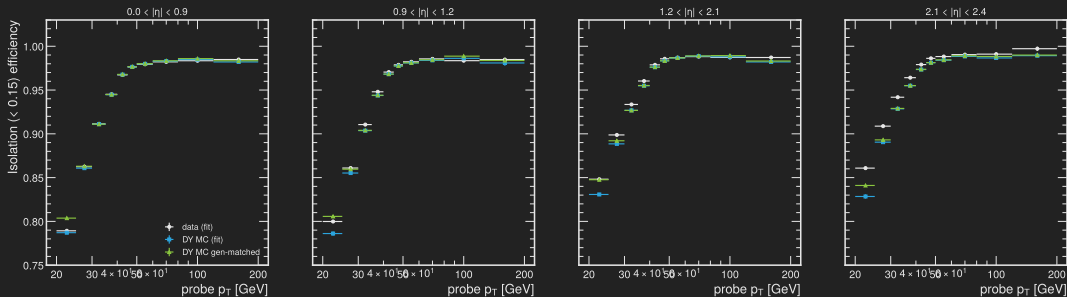
SS $\mu\mu$: prompt MC only; the rest is the non-prompt background (fake-factor closure region)



$e\mu$ "jets" (not cleaned): the excess sits in the bins with no real jet



$e\mu$: leading muon p_T



Isolation efficiency vs probe p_T , data and MC fits and gen-matched truth; the fit-vs-truth closure is -0.2% .